PROOF OF THE HASSETT–TSCHINKEL WEAK APPROXIMATION CONJECTURE

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Abstract. We prove the Hassett–Tschinkel conjecture that smooth geometrically rationally connected varieties over the function field of a complex curve satisfy weak approximation at all places, including places of bad reduction.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let B be a smooth projective curve with function field F, and call the closed points of B places. For a place b, let F_b be the fraction field of the completion $\widehat{\mathcal{O}}_{B,b}$ of the local ring $\mathcal{O}_{B,b}$. A variety X over F satisfies weak approximation if for any finite set S of places, the image of $X(F)$ under the diagonal map is dense in

$$\prod_{b\in S} X(F_b),$$

where each $X(F_b)$ carries the topology induced by the valuation topology of F_b. [GHS03, Theorem 1.2] proved that X has an F-point when X is smooth, projective, and geometrically rationally connected. Hassett and Tschinkel proposed the following conjecture.

Conjecture 1.1 ([HT06, Conjecture 2])**.** *Let B be a smooth projective curve with function field F. Then any smooth geometrically rationally connected variety over F satisfies weak approximation.*

We refer the reader to Definitions 2.1 and 2.5 for the definitions.

[HT06, Theorem 3] proved Conjecture 1.1 at places of good reduction. More precisely, let X be a smooth, proper, and geometrically rationally connected variety over F. A place b is a place of good reduction of X in the sense of [HT06, §2.2] if $X \times_F F_b$ extends to a smooth proper model over $\operatorname{Spec} \widehat{\mathcal{O}}_{B,b}$, which [HT06, §2.2] allows to be an algebraic space. Then for any finite set S of places of good reduction of X, the image of $X(F)$ is dense in $\prod_{b\in S} X(F_b)$. See also [Xu12, Conjecture 1.1], which formulates the part of Conjecture 1.1 left open by [HT06, Theorem 3] as weak approximation at places of bad reduction.

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We now list the known cases of Conjecture 1.1 that include places of bad reduction. Conjecture 1.1 holds for stably rational varieties [CTG04, Proposition 2.1], connected linear algebraic groups [CTG04, Théorème 4.1] and their homogeneous spaces [CTG04, Théorème 4.3], homogeneous space fibrations over varieties satisfying weak approximation (for example conic bundles over rational varieties) [CTG04, Théorème 4.4], and del Pezzo surfaces of degree at least four [CTG04, Corollaire 4.5, Remarque 4.6(a)]. See also [HT06, pp. 172–173].

The next group of results treats places of bad reduction through the smooth locus of the singular fibers of a projective model of X over B with smooth total space (see Definition 2.2).

(1) [HT08, Corollary 4] proved Conjecture 1.1 for cubic surfaces whose discriminant divisor on B is reduced. See also [Xu12, p. 754].
(2) [Kne13, Corollary 1.3] proved Conjecture 1.1 for del Pezzo surfaces of degree two whose discriminant divisor on B is reduced.
(3) [Xu12, Theorem 1.4] proved Conjecture 1.1 for del Pezzo surfaces of degree one admitting a model as in [Xu12, Definition 1.3].

The approach of [HT06, §3.2] to [HT06, Theorem 3] connects the intersection point of a given section and a fiber with a given point of the fiber by a chain of rational curves in the fiber, and then smooths the comb formed by the section and the chain. Sections of a projective model with smooth total space pass only through smooth points of the fibers [HT06, p. 172]. Thus at a place of bad reduction, the rational curves have to be contained in the smooth locus of the singular fiber. Indeed, the criterion [Xu12, Theorem 2.3] asks for a rational curve in the smooth locus of the singular fiber through any point and a general point, and [Has10, Theorem 2.12] constructs sections with given jets when the smooth locus of each fiber of such a model is strongly rationally connected in the sense of [Has10, Definition 2.10], that is, when through any point of the smooth locus of each fiber there passes a very free rational curve contained in the smooth locus of the fiber.

After resolving the singularities, a singular fiber often has more than one component, and to deform a section to one with the given jets, we have to find auxiliary rational curves in the fiber that correct the intersection numbers of the section with the components of the fiber [Xu12, p. 756]. In [Has10, §2], the singular fibers are described as the main technical challenge. So the method based on the smooth locus of the singular fibers depends on an analysis of the singular fibers case by case.

Global methods prove Conjecture 1.1 at all places for several classes of varieties.

(1) [HT09] proved Conjecture 1.1 for smooth hypersurfaces in $\mathbb{P}^n$ of low degree compared with n. See also [Has10, Theorem 3.4].
(2) [Has10, Theorem 4.7] proved Conjecture 1.1 at all places for smooth projective X under a rational simple connectedness hypothesis on X. We refer the reader to [BW25, §2.1] for a closely related definition of rationally simply connected varieties.
(3) [TZ19, Theorem 1.5] proved Conjecture 1.1 for smooth projective X at places of strong potentially good reduction in the sense of [TZ19, Definition 1.4], in particular for isotrivial families [TZ19, Theorem 1.7]. We refer the reader to [TZ19, p. 2] for the definition of isotrivial families.
(4) [Tia15, Theorem 1.2] proved Conjecture 1.1 at all places for smooth cubic hypersurfaces of dimension at least two.

Moreover, [RS09, Theorem 1.6] proved a local-to-global principle, which reduces weak approximation to a formal local problem near the fibers over the given places. The results of [HT09], [Has10], [TZ19], and [Tia15] depend on hypotheses on X itself. Rational simple connectedness, introduced in [Sta06] and developed in [dJHS11], asks that some spaces of rational curves on X be themselves rationally connected [BW25, p. 1166]. [TZ19, Definition 1.4] asks for a smooth proper model over the ring of integers of a cyclic extension of F_b such that the action of the Galois group extends to the model compatibly with the projection.

We now explain the recent work closest to this paper. [BW25, Theorem 1.4] proved that any smooth projective rationally simply connected variety over F satisfies the tight approximation property introduced in [BW21], which implies weak approximation [BW25, p. 1167]. The Alarcón–Forstnerič conjecture [AF25, Conjecture 9.1] predicts that any rationally connected smooth projective manifold is Oka-1. Towards [AF25, Conjecture 9.1], [DGX26, Theorem 1.7] proved that smooth projective morphisms with connected rationally connected fibers are algebraic Oka-1 maps, and [DGX26, Corollary 1.8] shows that a connected smooth projective complex manifold is algebraically Oka-1 in the sense of [FL25, Definition 1.5] if and only if it is rationally connected. The analytic input of this paper is taken from [DGX26].

We now explain why [DGX26, Theorem 1.7] does not yield weak approximation at places of bad reduction. Let $\pi : \mathcal{X} \to B$ be a model of X such that $\mathcal{X}$ is smooth and projective, let $\mathcal{X}^\circ$ be the smooth locus of π, and let $T_{\mathcal{X}/B}$ be the vertical tangent bundle, that is, the kernel of $d\pi$ on $\mathcal{X}^\circ$. In the projective model used in the proof of [DGX26, Theorem 1.7], the morphism π is smooth over an affine open subset $B\backslash\Sigma$ of B, where Σ is a finite set of places, and the singular fibers of π occur only over Σ. Accordingly, [DGX26, Theorem 1.7] contains no statement at places of bad reduction [DGX26, §1.4].

Let $D \geq 0$ be a divisor on B supported on the finitely many places where jets are given, let s be a smooth section of π with $s(B) \subset \mathcal{X}^\circ$, and let h be the algebraic section of [DGX26, Proposition 6.8]. Let $\mathcal{E}_s(-D)$ be the holomorphic vector bundle on B with underlying smooth bundle $s^*T_{\mathcal{X}/B}(-D)$ constructed in [DGX26, Lemma 4.2]. The proof in [DGX26, §7.2] uses a holomorphic vector bundle V on B with vanishing first cohomology together with a smooth isomorphism $V \to \mathcal{E}_s(-D)$ which is holomorphic near the places where jets are given and near Σ. There $V = h^*T_{\mathcal{X}/B}(-D)$, which satisfies

$$H^1(B, h^*T_{\mathcal{X}/B}(-D)) = 0$$

by [DGX26, Proposition 6.8], and the smooth isomorphism is obtained by [DGX26, Lemma 6.2] from the parallel transport $h^*T_{\mathcal{X}/B}(-D) \to \mathcal{E}_s(-D)$ along a homotopy of sections from h to s [DGX26, Lemma 6.1]. The homotopy is constructed by [DGX26, Lemma 2.4] over $B\backslash\Sigma$, where π is a fiber bundle with connected and simply connected fibers, and $s = h$ near the singular fibers. To prescribe a jet at a singular fiber $\pi^{-1}(b)$, the section s has to coincide near b with a given holomorphic local section u_b of π. A homotopy of sections with image in $\mathcal{X}^\circ$ restricts at b to a path in $\pi^{-1}(b) \cap \mathcal{X}^\circ$. Therefore, if $h(b)$ and $u_b(b)$ belong to different connected components of $\pi^{-1}(b) \cap \mathcal{X}^\circ$, then the homotopy from h to s does not exist.

For example, let $C = \{b\} \times \mathbb{P}^1 \subset B \times \mathbb{P}^1$ and let $\mu : \mathcal{X} \to B \times \mathbb{P}^1$ be the blow-up at a point of C, with exceptional curve E. Let $\pi = \mathrm{pr}_1 \circ \mu$, and let $\widetilde{C}$ be the strict transform of C. Then $\pi^*(b) = \widetilde{C} + E$, the curves $\widetilde{C}$ and E intersect at one point p, and

$$\pi^{-1}(b) \cap \mathcal{X}^\circ = (\widetilde{C}\backslash p) \sqcup (E\backslash p).$$

The strict transforms of two sections $B \times \{t_1\}$ and $B \times \{t_2\}$ with $t_1, t_2 \in \mathbb{P}^1$ can pass through different components of $\pi^{-1}(b) \cap \mathcal{X}^\circ$. Moreover, since $K_{\mathcal{X}} = \mu^*K_{B\times\mathbb{P}^1} + E$ [Har77, Proposition V.3.3], we have

$$-K_{\mathcal{X}/B} = \mu^* \,\mathrm{pr}_2^*\, \mathcal{O}_{\mathbb{P}^1}(2) - E,$$

and $T_{\mathcal{X}/B}$ is the line bundle on $\mathcal{X}^\circ$ corresponding to $-K_{\mathcal{X}/B}$, since $\Omega_{\mathcal{X}/B}$ is locally free on $\mathcal{X}^\circ$ [Har77, Proposition II.8.11, Proposition III.10.4]. Thus any smooth section s with $s(B) \subset \mathcal{X}^\circ$ satisfies

$$\deg s^*T_{\mathcal{X}/B} \equiv s_*[B] \cdot E \pmod 2.$$

Since $s(B) \cap \pi^{-1}(b) = \{s(b)\}$ and $s(B)$ is transversal to $\pi^{-1}(b)$ at $s(b)$ with sign $+1$ because $d\pi \circ ds = \mathrm{id}$, we have $s_*[B] \cdot E = 1$ if $s(b) \in E\backslash p$ and $s_*[B] \cdot E = 0$ if $s(b) \in \widetilde{C}\backslash p$. Therefore, for two smooth sections passing through different components of $\pi^{-1}(b) \cap \mathcal{X}^\circ$, the pullbacks of $T_{\mathcal{X}/B}$ are not even smoothly isomorphic. Here $X = \mathbb{P}^1_F$ satisfies weak approximation [CTG04, Proposition 2.1], so the disconnectedness of $\pi^{-1}(b) \cap \mathcal{X}^\circ$, which blocks the homotopy from h to s,

is a property of the model π and not of X. To repair the argument of [DGX26, §7.2], one needs an algebraic section through a given component of $\pi^{-1}(b)\cap\mathcal{X}^\circ$ for any singular fiber $\pi^{-1}(b)$. This is a weak form of [HT06, Conjecture 5], which predicts that for a model of X with smooth total space and smooth points $r_1,\dots,r_k$ of distinct fibers, there exists a section through $r_1,\dots,r_k$.

In this paper, we do not compare the section to which the approximation theorem [DGX26, Theorem 7.1] is applied with any algebraic section. Since two smooth complex vector bundles V_1, V_2 on B with $\operatorname{rank} V_1 = \operatorname{rank} V_2$ and $\deg V_1 = \deg V_2$ are smoothly isomorphic (see Lemma 2.8(3)), it suffices to make $\deg s^*T_{\mathcal{X}/B}$ large, and then to take a holomorphic vector bundle on B with vanishing first cohomology that does not depend on the geometry of π.

As of September 2026, Conjecture 1.1 at places of bad reduction is open in general. For example, it is open for del Pezzo surfaces of degree at most two [Dan26, p. 15], apart from special cases such as those of [Kne13, Corollary 1.3] and [Xu12, Theorem 1.4]. The goal of this paper is to prove Conjecture 1.1.

We have the following theorem.

Theorem 1.2. *Let B be a smooth projective curve with function field F. Then any smooth proper geometrically rationally connected variety over F satisfies weak approximation.*

Remark 1.3. Theorem 1.2 confirms Conjecture 1.1. Indeed, by [HT06, Remark 12], it suffices to prove Conjecture 1.1 for smooth projective varieties. Let X be a smooth geometrically rationally connected variety over F and $\overline{F}$ an algebraic closure of F. By [HT06, §2.1], there exists a smooth projective variety X' over F birational to X. The birational map

$$X\times_F\overline{F}\dashrightarrow X'\times_F\overline{F}$$

restricts to an isomorphism $h: U\to U'$ between dense open subsets. Since $\overline{F}$ is uncountable, two very general closed points of $X\times_F\overline{F}$ are contained in an irreducible rational curve by [Kol96, Proposition IV.3.6]. Let x_1' and x_2' be two very general closed points of $X'\times_F\overline{F}$. Then $h^{-1}(x_1')$ and $h^{-1}(x_2')$ are very general closed points of $X\times_F\overline{F}$, so they are contained in an irreducible rational curve $C\subset X\times_F\overline{F}$. Thus the closure of $h(C\cap U)$ in $X'\times_F\overline{F}$ is an irreducible rational curve containing x_1' and x_2'. Hence X' is geometrically rationally connected by [Kol96, Proposition IV.3.6], and X' satisfies weak approximation by Theorem 1.2.

By [HT06, §2.1] (see Lemma 2.4), Theorem 1.2 has the following geometric form in terms of models, where $\widehat{B}_b$ is the scheme of Definition 2.1 and $j_b^{N_b}$ denotes the N_b-jet of Definition 2.3.

Corollary 1.4. *Let B be a smooth projective curve with function field F, X a smooth proper geometrically rationally connected variety over F, and $\pi:\mathcal{X}\to B$ a model of X. Let S be a finite set of closed points of B. For any $b\in S$, let $\hat{x}_b:\widehat{B}_b\to\mathcal{X}$ be a morphism over B and $N_b\geq 0$ an integer. Then there exists a section $\sigma:B\to\mathcal{X}$ of π such that $j_b^{N_b}\sigma = j_b^{N_b}\hat{x}_b$ for any $b\in S$.*

Sketch of the proof. By [HT06, §2.1] and Lemma 2.4, Theorem 1.2 follows from Proposition 5.2, the case of Corollary 1.4 for the model $\pi:\mathcal{X}\to B$ of Construction-Proposition 3.2, which has smooth projective total space and is smooth over a non-empty Zariski open subset $\Omega\subset B$. Proposition 5.2 follows from Theorem 2.11, a case of [DGX26, Theorem 7.1], and Chow's theorem [Cho49, Theorem V], applied to data constructed in three steps.

(1) Construction-Proposition 3.3 and Proposition 3.4 construct a smooth section s_0 of π which is holomorphic near $S'=S\cup(B\backslash\Omega)$ and satisfies $s_0(B)\subset\mathcal{X}^\circ$ and $j_b^{N_b}s_0 = j_b^{N_b}\hat{x}_b$ for any $b\in S$.

(2) Proposition 4.1 glues k copies of a very free rational curve in a fiber over Ω into s_0 and obtains sections s_k equal to s_0 near S' with

$$\deg s_k^*T_{\mathcal{X}/B} = \deg s_0^*T_{\mathcal{X}/B} + ka$$

for a positive integer a independent of k.

(3) For $s = s_k$ with k large, Proposition 5.1 constructs a holomorphic vector bundle $\mathcal{V}$ with $H^1(B, \mathcal{V}) = 0$ and a smooth isomorphism $\phi : \mathcal{V} \to \mathcal{E}_s(-D)$ (Construction 2.10) holomorphic near S'.

Smooth complex vector bundles on B of equal rank and degree are isomorphic (Lemma 2.8(3)), so the bundle data in (3) exist once $\deg s^*T_{\mathcal{X}/B}$ is large. Only (2) and the choice of $\mathcal{V}$ in (3) are new, and the fibers over $B\backslash\Omega$ enter only through the germs of s_0 near S'. The proof is a pure existence argument and does not bound the degree of $\sigma(B)$.

By GAGA [Ser56, Théorèmes 1, 3] and the Riemann–Roch theorem [Har77, Appendix A, Theorem 4.1], a holomorphic vector bundle $\mathcal{V}$ of rank $d = \dim X$ on B with $H^1(B, \mathcal{V}) = 0$ satisfies $\deg \mathcal{V} \geq d(g-1)$, where g is the genus of B (by Lemma 2.8(1) applied to $\det \mathcal{V}$, $\deg \mathcal{V}$ is the degree of the algebraic vector bundle given by GAGA), so Theorem 2.11 can be used for s only if $\deg s^*T_{\mathcal{X}/B} \geq d(\deg D + g - 1)$. In [DGX26, §7.2], the inequality follows from a homotopy of sections from an algebraic section to s, and the example of the blow-up of $B \times \mathbb{P}^1$ at a point of a fiber, given before Theorem 1.2, shows that such a homotopy need not exist when s is given near a singular fiber. Here $\deg s_0^*T_{\mathcal{X}/B}$ is not controlled, since over a 2-cell of B an extension of a section is determined only up to an element of π_2 of a fiber. For $c \in \Omega$, the fiber $\pi^{-1}(c)$ is rationally connected, so there exists a very free morphism $f : \mathbb{P}^1 \to \pi^{-1}(c)$ [DGX26, Theorem 3.1(i)] (see also [Kol00, Definition–Theorem 2.1(5)]), and

$$\langle c_1(T_{\pi^{-1}(c)}), f_*[\mathbb{P}^1] \rangle = \deg f^*T_{\pi^{-1}(c)} \geq d.$$

Proposition 4.1 glues k copies of f into s_0 over small disks in Ω away from S', which adds $kf_*[\mathbb{P}^1]$ to the class $s_{0*}[B] \in H_2(\mathcal{X}^\circ; \mathbb{Z})$, whose pairing with $c_1(T_{\mathcal{X}/B})$ is $\deg s_0^*T_{\mathcal{X}/B}$.

The bundle $\mathcal{E}_s(-D)$ itself cannot serve as $\mathcal{V}$ in general: rank and degree do not determine H^1 of a holomorphic vector bundle (see the example after Theorem 2.11), and the holomorphic structure of $\mathcal{E}_s(-D)$ is not controlled away from the locus where s is holomorphic. Instead, Proposition 5.1 takes for $\mathcal{V}$ a direct sum of d line bundles $\mathcal{O}_B(qb_0)$ and $\mathcal{O}_B((q+1)b_0)$ with $\deg \mathcal{V} = \deg \mathcal{E}_s(-D)$, where $b_0 \in B$ is a closed point. Then $H^1(B, \mathcal{V}) = 0$ once $q > 2g - 2$, which follows from the bound $\deg s^*T_{\mathcal{X}/B} \geq d \deg D + 2gd$ in Proposition 5.1, valid for k large, and [DGX26, Lemma 6.2] modifies a smooth isomorphism $\mathcal{V} \to \mathcal{E}_s(-D)$ so that it is holomorphic near S'.

Theorem 2.11 has no hypothesis on the fibers of π over $B\backslash\Omega$, since its proof replaces s by rational disks in fibers of π, given by the families in Definition 2.9, only over a compact subset of Ω, and near $B\backslash\Omega$ the section σ in Theorem 2.11 is a vertical perturbation of s [DGX26, proof of Theorem 7.1, Steps 1, 2, and 6]. The rational connectedness of the fibers over Ω is used four times: it implies their simple connectedness [DGX26, Theorem 3.1(ii)] (see also [Kol00, Proposition 2.3]), which is used in (1) together with [DGX26, Lemma 2.3]; it implies that π has a section [GHS03, Theorem 1.1], which is used in (1) for the germs of s_0 at the points of $B\backslash\Omega$ outside S; it implies the existence of the very free curves in (2); and it implies the relative first-jet sphere-family property of π over Ω [DGX26, Proposition 3.5], which is a hypothesis of Theorem 2.11.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce models and jets, and recall an approximation theorem for holomorphic sections [DGX26, Theorem 7.1].

In Section 3 we choose the model $\pi : \mathcal{X} \to B$ and construct the smooth section s_0 of step (1) of the sketch of the proof (Proposition 3.4).

In Section 4 we insert very free rational curves into s_0, which is step (2) (Proposition 4.1).

In Section 5 we construct the bundle data of step (3) (Proposition 5.1) and deduce Theorem 1.2 and Corollary 1.4 from [DGX26, Theorem 7.1].

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2. Preliminaries

2.1. Weak approximation and models.

Definition 2.1. Let B be a smooth projective curve with function field F. For a closed point $b \in B$, we denote by $\widehat{\mathcal{O}}_{B,b}$ the completion of the local ring $\mathcal{O}_{B,b}$, by $\mathfrak{m}_b$ the maximal ideal of $\widehat{\mathcal{O}}_{B,b}$, by F_b the fraction field of $\widehat{\mathcal{O}}_{B,b}$, and by $\widehat{B}_b$ the scheme $\operatorname{Spec} \widehat{\mathcal{O}}_{B,b}$. For a variety X over F, we equip $X(F_b)$ with the topology induced by the $\mathfrak{m}_b$-adic topology of F_b via affine open subsets of X. We say that X satisfies *weak approximation* if for any finite set S of closed points of B, the image of the diagonal map $X(F) \to \prod_{b\in S} X(F_b)$ is dense in $\prod_{b\in S} X(F_b)$ with the product topology.

For $S = \emptyset$, the condition in Definition 2.1 means $X(F) \neq \emptyset$. For any local parameter t_b of B at b, we have $F_b \cong \mathbb{C}((t_b))$ (Cohen structure theorem [Har77, Theorem I.5.5A]).

Definition 2.2. Let B be a smooth projective curve with function field F and X a proper variety over F. A *model* of X is a proper morphism $\pi : \mathcal{X} \to B$ from a variety $\mathcal{X}$ together with an isomorphism over F between X and the generic fiber of π.

Since $\mathcal{X}$ is a variety dominating the smooth curve B, π is flat [Har77, Proposition III.9.7], so a model in Definition 2.2 is a proper model in the sense of [HT06, §2.1].

Definition 2.3. Let B be a smooth projective curve, $b \in B$ a closed point, and $N \geq 0$ an integer.

(1) Let $\mathcal{X}$ be a variety and $h_1, h_2 : \widehat{B}_b \to \mathcal{X}$ two morphisms. We write $j_b^N h_1 = j_b^N h_2$ if h_1 and h_2 coincide on the closed subscheme of $\widehat{B}_b$ defined by $\mathfrak{m}_b^{N+1}$.
(2) Let $\mathcal{X}$ be a complex manifold and h_1, h_2 two holomorphic maps from a neighborhood of b in B to $\mathcal{X}$. We write $j_b^N h_1 = j_b^N h_2$ if $h_1(b) = h_2(b)$ and, for some (equivalently, any; see [DGX26, §1.1]) local coordinate t of B centered at b and holomorphic chart ψ of $\mathcal{X}$ at $h_1(b)$, each component of $\psi \circ h_1 - \psi \circ h_2$ is divisible by t^{N+1}.

A morphism or a holomorphic map from a neighborhood of b in B to a variety $\mathcal{X}$ induces a morphism $\widehat{B}_b \to \mathcal{X}$, and we apply (1) to the induced morphisms.

Lemma 2.4 ([HT06, §2.1])**.** *Let B be a smooth projective curve with function field F, X a smooth proper variety over F, $\pi : \mathcal{X} \to B$ a model of X, and $b \in B$ a closed point. Then any $x \in X(F_b)$ extends uniquely to a morphism $\hat{x} : \widehat{B}_b \to \mathcal{X}$ over B. Moreover, the sets $\{y \in X(F_b) \mid j_b^N \hat{y} = j_b^N \hat{x}\}$ for integers $N \geq 0$ form a basis of open neighborhoods of x in $X(F_b)$, where $\hat{y} : \widehat{B}_b \to \mathcal{X}$ denotes the extension of y.*

The existence and the uniqueness of $\hat{x}$ in Lemma 2.4 follow from the valuative criterion of properness [Har77, Theorem II.4.7].

2.2. Rationally connected varieties.

Definition 2.5. We say that a variety Y over an algebraically closed field is *rationally connected* [Cam92, KMM92] if there exists a family of proper algebraic curves $g : U \to Z$ whose geometric fibers are irreducible rational curves with cycle morphism $u : U \to Y$ such that the morphism $u^{(2)} : U \times_Z U \to Y \times Y$ induced by u is dominant [Kol96, Definition IV.3.2]. Let B be a smooth projective curve with function field F and $\overline{F}$ an algebraic closure of F. We say that a variety X over F is *geometrically rationally connected* if $X \times_F \overline{F}$ is rationally connected. For a smooth variety Y, we say that a morphism $f : \mathbb{P}^1 \to Y$ is *very free* if f^*T_Y is ample [HT06, Definition 15].

Lemma 2.6. *Let Y be a rationally connected smooth projective variety. Then any two closed points of Y are contained in the image of a morphism $\mathbb{P}^1 \to Y$. In particular, Y is rationally connected in the sense of* [DGX26, §1.1].

Proof. Since the characteristic is zero, Y is separably rationally connected [Kol96, Proposition IV.3.3, (3.3.1)]. Let $y_1, y_2 \in Y$ be two closed points. If $y_1 \neq y_2$, then [Kol96, Theorem IV.3.9] with $m = 2$ and the points y_1, y_2 implies that there exists a morphism $f : \mathbb{P}^1 \to Y$ with $y_1, y_2 \in f(\mathbb{P}^1)$. If $y_1 = y_2$, then [Kol96, Theorem IV.3.9] with $m = 1$ and the point y_1 implies that there exists a morphism $f : \mathbb{P}^1 \to Y$ with $y_1 \in f(\mathbb{P}^1)$.

By [DGX26, §1.1], Y is rationally connected in the sense of [DGX26, §1.1] if there exists a non-empty Zariski open subset $U \subset Y$ such that any two points of U are contained in the image of some morphism $\mathbb{P}^1 \to Y$. Thus $U = Y$ satisfies the condition in [DGX26, §1.1]. □

2.3. **An approximation theorem for holomorphic sections.**

Definition 2.7. Let B be a smooth projective curve. For a smooth complex vector bundle E on B, we call $\deg E := \langle c_1(E), [B] \rangle$ the *degree* of E, where $[B]$ is the fundamental class of B with the orientation induced by the complex structure.

The sections s of π constructed in Sections 3 and 4 are only smooth, so the pullback $s^*T_{\mathcal{X}/B}$ of the vertical tangent bundle (see Construction 2.10) is only a smooth complex vector bundle, and $\mathcal{E}_s$ denotes the holomorphic structure on it defined in Construction 2.10. By Lemma 2.8(3), Section 5 uses only the rank and the degree of $s^*T_{\mathcal{X}/B}$.

Lemma 2.8. *Let B be a smooth projective curve.*

(1) For any divisor D on B, the underlying smooth complex line bundle of $\mathcal{O}_B(D)$ has degree $\deg D$.

(2) For smooth complex vector bundles E_1, E_2 and a smooth complex line bundle L on B, we have $\deg(E_1 \oplus E_2) = \deg E_1 + \deg E_2$ and $\deg(E_1 \otimes L) = \deg E_1 + \operatorname{rank} E_1 \cdot \deg L$.

(3) Smooth complex vector bundles on B are determined up to smooth isomorphism by their rank and their degree.

Proof. (1) This is [GH78, p. 144].

(2) By [GH78, p. 408],

$$c_1(E_1 \oplus E_2) = c_1(E_1) + c_1(E_2) \quad \text{and} \quad c_1(E_1 \otimes L) = c_1(E_1) + \operatorname{rank} E_1 \cdot c_1(L).$$

Pairing with $[B]$ implies (2).

(3) Let E be a smooth complex vector bundle of rank $r \geq 2$ on B. There exists a smooth section of E transverse to the zero section (by the transversality theorem, a generic smooth section is), and such a section has no zeros because the zero set has real codimension $2r > 2 = \dim_{\mathbb{R}} B$. The orthogonal complement of the image of a nowhere vanishing smooth section with respect to a Hermitian metric on E is a smooth complex vector bundle E' of rank $r - 1$ with $E \cong E' \oplus \underline{\mathbb{C}}$, where $\underline{\mathbb{C}}$ denotes the trivial smooth complex line bundle on B. By induction on r, any smooth complex vector bundle E of rank $r \geq 1$ on B is smoothly isomorphic to $L \oplus \underline{\mathbb{C}}^{\oplus(r-1)}$ for some smooth complex line bundle L on B, and $\deg L = \deg E$ by (2) since $\deg \underline{\mathbb{C}} = 0$. Smooth complex line bundles on B are determined up to smooth isomorphism by their first Chern classes [GH78, p. 140], and pairing with $[B]$ induces an isomorphism $H^2(B, \mathbb{Z}) \cong \mathbb{Z}$ (Poincaré duality for the compact connected oriented surface B). Thus two smooth complex line bundles L_1, L_2 on B with $\deg L_1 = \deg L_2$ are smoothly isomorphic, and (3) follows. □

Definition 2.9. Let $\pi : \mathcal{X} \to \Omega$ be a holomorphic submersion between complex manifolds and $T_{\mathcal{X}/\Omega} := \ker d\pi$ the vertical tangent bundle. Fix an affine coordinate w on $\mathbb{P}^1$ centered at 0. We say that π has the *relative first-jet sphere-family property* if for any $(x, v) \in T_{\mathcal{X}/\Omega}$, there exist an open neighborhood $\mathcal{U}$ of (x, v) in $T_{\mathcal{X}/\Omega}$ and a holomorphic map $G : \mathbb{P}^1 \times \mathcal{U} \to \mathcal{X}$ such that $\pi(G(w; x', v')) = \pi(x')$, $G(0; x', v') = x'$, and $\partial_w G(0; x', v') = v'$ for any $(x', v') \in \mathcal{U}$ [DGX26, Definition 1.3].

For $v \neq 0$, the property in Definition 2.9 asks for a rational curve in the fiber of π through x with tangent vector v at x, depending holomorphically on (x, v). It is the hypothesis of [DGX26, Lemma 5.1], which uses that G depends jointly holomorphically on (x', v'). The existence of $G(\cdot; x, v)$ for each (x, v) separately would not suffice.

Theorem 2.11 requires the property in Definition 2.9 only for $\pi^{-1}(\Omega) \to \Omega$. In this paper, the property follows from [DGX26, Proposition 3.5], which considers smooth projective morphisms of complex manifolds with rationally connected fibers, together with Lemma 2.6.

Construction 2.10 uses a vertical local addition as in [DGX26, Lemma 4.1]. It is a smooth map defined near the zero vectors of $T_{\mathcal{X}/B}$, with values in the fiber of π through the base point, holomorphic in the vector variable, sending the zero vector at x to x, and with derivative in the vector variable at the zero vector equal to the identity. By [DGX26, §4.1], the constructions of [DGX26, Lemma 4.1] and [DGX26, Lemma 4.2] also work near the compact image of a section of a holomorphic map contained in the submersion locus.

Construction 2.10 ([DGX26, Lemma 4.2])**.** Let B be a smooth projective curve, $\mathcal{X}$ a complex manifold, $\pi : \mathcal{X} \to B$ a holomorphic map, $\mathcal{X}^\circ \subset \mathcal{X}$ the submersion locus of π, and $T_{\mathcal{X}/B} := \ker(d\pi|_{\mathcal{X}^\circ})$ the vertical tangent bundle. Let $s : B \to \mathcal{X}$ be a smooth section of π with $s(B) \subset \mathcal{X}^\circ$, and fix a vertical local addition near $s(B)$ as in [DGX26, Lemma 4.1]. We denote by $\mathcal{E}_s$ the holomorphic vector bundle on B whose underlying smooth complex vector bundle is $s^*T_{\mathcal{X}/B}$ and whose holomorphic structure is the one constructed in [DGX26, Lemma 4.2]. For a divisor D on B, we write $\mathcal{E}_s(-D) := \mathcal{E}_s \otimes \mathcal{O}_B(-D)$.

If s is not holomorphic, then $\mathcal{E}_s$ may depend on the vertical local addition (see the paragraph before [DGX26, Lemma 4.3]). On any open subset of B on which s is holomorphic, the holomorphic structure of $\mathcal{E}_s$ is the natural holomorphic structure of $s^*T_{\mathcal{X}/B}$ [DGX26, Lemma 4.3].

Theorem 2.11 is the case of [DGX26, Theorem 7.1] in which the compact set is empty, with the compact Riemann surface written as the smooth projective curve B. The conclusion of [DGX26, Theorem 7.1] also contains a uniform approximation on the compact set, which this paper does not use: by Lemma 2.4, weak approximation depends only on the jets at the points of S.

Theorem 2.11 ([DGX26, Theorem 7.1])**.** *Let B be a smooth projective curve, $\mathcal{X}$ a complex manifold, and $\pi : \mathcal{X} \to B$ a proper holomorphic map. Let $\Omega \subset B$ be an open subset such that $\pi^{-1}(\Omega) \to \Omega$ is a holomorphic submersion with the relative first-jet sphere-family property. Let $s : B \to \mathcal{X}$ be a smooth section of π with $s(B)$ contained in the submersion locus $\mathcal{X}^\circ$ of π, and let $\mathcal{E}_s$ be as in Construction 2.10. Let $S \subset B$ be a finite subset and $N_b \geq 0$ an integer for any $b \in S$. Let $D := \sum_{b\in S}(N_b + 1)b$. Assume that there exist an open subset $U \subset B$ containing $S \cup (B\backslash\Omega)$, a holomorphic vector bundle $\mathcal{V}$ on B with $H^1(B, \mathcal{V}) = 0$, and a smooth complex vector bundle isomorphism $\phi : \mathcal{V} \to \mathcal{E}_s(-D)$ such that s and ϕ are holomorphic on U. Then there exists a holomorphic section $\sigma : B \to \mathcal{X}$ of π such that $j_b^{N_b}\sigma = j_b^{N_b}s$ for any $b \in S$.*

Example 2.12. The hypothesis of Theorem 2.11 asks for a smooth isomorphism from some $\mathcal{V}$ with $H^1(B, \mathcal{V}) = 0$, and not for $H^1(B, \mathcal{E}_s(-D)) = 0$. Indeed, the holomorphic structure of $\mathcal{E}_s$ depends on s off the locus where s is holomorphic and on the vertical local addition, and holomorphic structures on a smooth complex vector bundle are not determined by the rank and the degree: for any integer $m \geq -2$, the bundles

$$\mathcal{O}(-5) \oplus \mathcal{O}(m + 5) \quad \text{and} \quad \mathcal{O}(\lfloor m/2 \rfloor) \oplus \mathcal{O}(\lceil m/2 \rceil)$$

on $\mathbb{P}^1$ have rank 2 and degree m (Lemma 2.8(1)(2)), hence are smoothly isomorphic by Lemma 2.8(3), while

$$h^1(\mathbb{P}^1, \mathcal{O}(-5) \oplus \mathcal{O}(m + 5)) \geq h^1(\mathbb{P}^1, \mathcal{O}(-5)) = 4$$

and

$$H^1(\mathbb{P}^1, \mathcal{O}(\lfloor m/2 \rfloor) \oplus \mathcal{O}(\lceil m/2 \rceil)) = 0$$

[Har77, Theorem III.5.1] (for coherent sheaves on $\mathbb{P}^1$, holomorphic and algebraic cohomology coincide by [Har77, Appendix B, Theorem 2.1]). Thus the rank and the degree of a holomorphic vector bundle $\mathcal{W}$ on $\mathbb{P}^1$ do not determine whether $H^1(\mathbb{P}^1, \mathcal{W}) = 0$.

3. Smooth sections with given germs

In this section, we construct a smooth section of a smooth projective model of the variety X in Set-up 3.1 whose image is contained in the smooth locus of the model, which is holomorphic near the singular fibers, and which has the given N_b-jets at the points of S.

Set-up 3.1. Let B, g, F, X, d, S, x_b, N_b, and D be as follows.

(1) B is a smooth projective curve of genus g with function field F.
(2) X is a smooth projective geometrically rationally connected variety over F of dimension $d \geq 1$.
(3) S is a finite set of closed points of B, and for any $b \in S$, x_b is a point of $X(F_b)$ and $N_b \geq 0$ is an integer.
(4) $D := \sum_{b \in S}(N_b + 1)b$.

In (2), the conditions that X is projective and $d \geq 1$ cause no loss of generality for Theorem 1.2 (see Section 5).

Construction-Proposition 3.2. Notation and conditions as in Set-up 3.1(1) and (2). Then there exist a model $\pi : \mathcal{X} \to B$ of X with $\mathcal{X}$ a smooth projective variety and a non-empty Zariski open subset $\Omega \subset B$ such that π is smooth over Ω and $\pi^{-1}(c)$ is a rationally connected smooth projective variety for any closed point $c \in \Omega$. We fix such $\pi : \mathcal{X} \to B$ and Ω, and denote by $\mathcal{X}^\circ \subset \mathcal{X}$ the open subset where π is smooth and by $T_{\mathcal{X}/B} := \ker(d\pi|_{\mathcal{X}^\circ})$ the vertical tangent bundle.

Proof. Fix a closed embedding $X \subset \mathbb{P}^n_F$, and let $\overline{\mathcal{X}}$ be the closure of X in $\mathbb{P}^n_B$, where $\mathbb{P}^n_F$ is the generic fiber of $\mathbb{P}^n_B \to B$. Then $\overline{\mathcal{X}}$ is a projective variety, and the generic fiber of the induced morphism $\overline{\mathcal{X}} \to B$ is X. Let $h : \mathcal{X} \to \overline{\mathcal{X}}$ be a projective resolution of singularities which is an isomorphism over the regular locus of $\overline{\mathcal{X}}$ [Hir64], [Kol07, Theorem 3.36], and let $\pi : \mathcal{X} \to B$ be the induced morphism. Since X is smooth over F, the local rings of $\overline{\mathcal{X}}$ at the points of X are regular. Thus the generic fiber of π is X, and $\pi : \mathcal{X} \to B$ is a model of X with $\mathcal{X}$ a smooth projective variety.

Since X is geometrically rationally connected, X is geometrically integral, so $H^0(X, \mathcal{O}_X) = F$ (by flat base change [Har77, Proposition III.9.3] and [Har77, Exercise II.4.5(d)] applied to $X \times_F \overline{F}$). Thus $\pi_*\mathcal{O}_{\mathcal{X}}$ is a coherent $\mathcal{O}_B$-algebra contained in the constant sheaf F. Since B is normal, $\pi_*\mathcal{O}_{\mathcal{X}} = \mathcal{O}_B$, and any fiber of π is connected [Har77, Corollary III.11.3]. Since X is smooth over F and π is proper, there exists a non-empty Zariski open subset $\Omega_1 \subset B$ such that π is smooth over Ω_1. In particular, $\pi^{-1}(c)$ is a smooth projective variety for any closed point $c \in \Omega_1$.

Since $X \times_F \overline{F}$ is rationally connected and the characteristic is zero, $X \times_F \overline{F}$ is separably rationally connected [Kol96, Proposition IV.3.3, (3.3.1)]. Thus there exists a very free morphism $f : \mathbb{P}^1_{\overline{F}} \to X \times_F \overline{F}$ [Kol96, Theorem IV.3.7], and f is the base change of a morphism f' from $\mathbb{P}^1_{F'}$ to $X \times_F F'$ for some finite extension F' of F (by the standard limit argument). Let $\beta : B' \to B$ be the normalization of B in F'. Then, again by the standard limit argument, there exist a non-empty Zariski open subset $V' \subset \beta^{-1}(\Omega_1)$ and a morphism $f_{V'} : \mathbb{P}^1_{V'} \to \mathcal{X} \times_B V'$ over V' whose generic fiber is f'. For a closed point $c' \in V'$, we denote by $f_{c'} : \mathbb{P}^1 \to \pi^{-1}(\beta(c'))$ the fiber of $f_{V'}$ over c'. Since any vector bundle on $\mathbb{P}^1$ over an algebraically closed field is a direct sum of line bundles [Har77, Exercise V.2.6] (for $\mathbb{P}^1_{\mathbb{C}}$, see also [Gro57, Théorème 2.1]), a morphism $f_0 : \mathbb{P}^1 \to Y$ to a smooth variety Y is very free if and only if

$$H^1(\mathbb{P}^1, f_0^* T_Y(-2)) = 0.$$

Applying the criterion to f (cohomology is compatible with the flat base change from F' to $\overline{F}$) and the semicontinuity theorem [Har77, Theorem III.12.8] to the pullback of $T_{\mathcal{X}/B}$ by $\mathbb{P}^1_{V'} \to \mathcal{X}$

twisted by $\mathcal{O}_{\mathbb{P}^1}(-2)$, possibly shrinking V', we may assume that $f_{c'}$ is very free for any closed point $c' \in V'$. Let

$$\Omega := B \backslash \beta(B' \backslash V').$$

Since β is finite and V' is non-empty, Ω is a non-empty Zariski open subset of B, and $\Omega \subset \Omega_1$. For any closed point $c \in \Omega$, we have $\beta^{-1}(c) \subset V'$, so $\pi^{-1}(c)$ contains the image of a very free morphism $\mathbb{P}^1 \to \pi^{-1}(c)$. Thus $\pi^{-1}(c)$ is separably rationally connected [Kol96, Theorem IV.3.7], hence rationally connected [Kol96, Proposition IV.3.3, (3.3.1)].

Since $\mathcal{X}$ and B are smooth, a closed point of $\mathcal{X}$ is a point where π is smooth if and only if the holomorphic map π is a submersion there [Har77, Proposition III.10.4] (applied to the restriction of π to an open neighborhood of the point, since the loci where π is smooth and where its tangent map is surjective are open). Thus $\mathcal{X}^\circ$ is the submersion locus of π in Construction 2.10. Moreover, Ω is also open in the complex topology of B. □

We make no assumption on the fibers of π over $B\backslash\Omega$, which may be singular, non-reduced, or reducible. Near the fibers over $B\backslash\Omega$, we use the smoothness of $\mathcal{X}$: Theorem 2.11 uses that $\mathcal{X}$ is a complex manifold, and Construction-Proposition 3.3 uses that the local ring of $\mathcal{X}$ at the center of a formal section is regular.

The next construction shows that the fibers over $B\backslash\Omega$ do not restrict the position of the jets: the center of a formal section is always a point where π is a submersion (for sections of a model with smooth total space, this is observed in [HT06, p. 172]), so a formal section can be truncated to a holomorphic local section. At a point of $B\backslash\Omega$ outside S, Theorem 2.11 also asks for a holomorphic local section of π.

Construction-Proposition 3.3. Notation and conditions as in Set-up 3.1 and Construction-Proposition 3.2. Let S' be the union of S and $B\backslash\Omega$, and for any $b \in S$, let $\hat{x}_b : \widehat{B}_b \to \mathcal{X}$ be the extension of x_b (Lemma 2.4). Then there exist pairwise disjoint closed coordinate disks $\Delta_b \subset B$ centered at b and holomorphic sections u_b of π over open neighborhoods of Δ_b for $b \in S'$, such that $u_b(\Delta_b) \subset \mathcal{X}^\circ$ for any $b \in S'$ and $j_b^{N_b} u_b = j_b^{N_b} \hat{x}_b$ for any $b \in S$. We fix such Δ_b and u_b.

Proof. Since S and $B\backslash\Omega$ are finite, S' is finite. For any $b \in S'$, let $t_b \in \mathcal{O}_{B,b}$ be a uniformizer. Then t_b is a holomorphic local coordinate of B on a neighborhood of b with $t_b(b) = 0$, and $\widehat{\mathcal{O}}_{B,b} = \mathbb{C}[[t_b]]$.

Fix $b \in S$. Let $z_b \in \mathcal{X}$ be the image of the closed point of $\widehat{B}_b$ under $\hat{x}_b$, and let $\mathfrak{m}_{z_b}$ be the maximal ideal of $\mathcal{O}_{\mathcal{X},z_b}$. Since $\pi \circ \hat{x}_b : \widehat{B}_b \to B$ is the canonical morphism, the local homomorphism

$$\hat{x}_b^\# : \mathcal{O}_{\mathcal{X},z_b} \to \widehat{\mathcal{O}}_{B,b}$$

maps $\pi^* t_b$ to t_b and $\mathfrak{m}_{z_b}^2$ into $\mathfrak{m}_b^2$. Since $t_b \notin \mathfrak{m}_b^2$, we have $\pi^* t_b \notin \mathfrak{m}_{z_b}^2$. Thus $d(\pi^* t_b)$ does not vanish at z_b, that is, $d\pi_{z_b} \neq 0$. Since B is a curve, π is a submersion at z_b, that is, $z_b \in \mathcal{X}^\circ$.

Hence there exist an open neighborhood $W_b \subset \mathcal{X}^\circ$ of z_b and holomorphic functions $w_1, \dots, w_d$ on W_b vanishing at z_b such that $(\pi^* t_b, w_1, \dots, w_d)$ is a holomorphic coordinate system on W_b. Since the completion of $\mathcal{O}_{\mathcal{X},z_b}$ coincides with the completion of the analytic local ring of $\mathcal{X}$ at z_b [Ser56, Proposition 3], both completions are $\mathbb{C}[[\pi^* t_b, w_1, \dots, w_d]]$, and $\hat{x}_b^\#$ extends to a local homomorphism from $\mathbb{C}[[\pi^* t_b, w_1, \dots, w_d]]$ to $\mathbb{C}[[t_b]]$. Write

$$\hat{x}_b^\#(w_i) = \sum_{k \geq 1} a_{i,k} t_b^k, \quad a_{i,k} \in \mathbb{C}, \quad 1 \leq i \leq d.$$

Then there exists an open neighborhood V_b of b such that, for any $c \in V_b$, the point $u_b(c) \in W_b$ with coordinates

$$\pi^* t_b = t_b(c), \quad w_i = \sum_{k=1}^{N_b} a_{i,k} t_b(c)^k, \quad 1 \leq i \leq d,$$

is defined. Thus $u_b : V_b \to W_b \subset \mathcal{X}^\circ$ is a holomorphic section of π over V_b. The morphism $\widehat{B}_b \to \mathcal{X}$ induced by u_b corresponds to the local homomorphism

$$u_b^\# : \mathbb{C}[[\pi^* t_b, w_1, \dots, w_d]] \to \mathbb{C}[[t_b]]$$

which maps $\pi^* t_b$ to t_b and w_i to $\sum_{k=1}^{N_b} a_{i,k} t_b^k$. Thus $u_b^\#$ and $\hat{x}_b^\#$ coincide modulo $t_b^{N_b+1}$, so $j_b^{N_b} u_b = j_b^{N_b} \hat{x}_b$.

Now assume that $S' \backslash S \neq \emptyset$. Since $\pi^{-1}(c)$ is rationally connected for any closed point $c \in \Omega$, the general fiber of π is rationally connected. By [GHS03, Theorem 1.1], there exists a section $\sigma : B \to \mathcal{X}$ of π. Since $\pi \circ \sigma = \mathrm{id}_B$, we have $d\pi \circ d\sigma = \mathrm{id}_{T_B}$, so $d\pi$ is surjective at any point of $\sigma(B)$, and $\sigma(B) \subset \mathcal{X}^\circ$. For any $b \in S' \backslash S$, let V_b be an open neighborhood of b on which t_b is a local coordinate, and let $u_b := \sigma|_{V_b}$.

Thus for any $b \in S'$, u_b is a holomorphic section of π over V_b with $u_b(V_b) \subset \mathcal{X}^\circ$. Since S' is finite, there exist real numbers $r_b > 0$ for $b \in S'$ such that the closed coordinate disks

$$\Delta_b := \{|t_b| \le r_b\}$$

are contained in V_b and pairwise disjoint. Then Δ_b and u_b satisfy the conditions. □

The next proposition only uses the fibers over Ω, which are connected and simply connected.

Proposition 3.4. *Notation as in Construction-Proposition 3.3. Then there exists a smooth section $s_0 : B \to \mathcal{X}$ of π such that $s_0(B) \subset \mathcal{X}^\circ$ and $s_0|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$.*

Proof. For any $b \in S'$, write $\Delta_b = \{|t_b| \le r_b\}$, where t_b is a holomorphic local coordinate of B centered at b and $r_b > 0$ is a real number. Since the disks Δ_b are compact and pairwise disjoint, there exist real numbers $0 < r_b^- < r_b < r_b^+$ for $b \in S'$ such that the closed coordinate disks

$$\Delta_b^- := \{|t_b| \le r_b^-\} \quad \text{and} \quad \Delta_b^+ := \{|t_b| \le r_b^+\}$$

satisfy the following: Δ_b^+ is contained in the domain of u_b for any $b \in S'$, and Δ_b^+ for $b \in S'$ are pairwise disjoint. Let

$$M_1 := B \backslash \bigcup_{b \in S'} \operatorname{int} \Delta_b^- \quad \text{and} \quad M_0 := \bigcup_{b \in S'} (\Delta_b^+ \backslash \operatorname{int} \Delta_b).$$

Since $B \backslash \Omega \subset S'$, $b \in \operatorname{int} \Delta_b^-$ for any $b \in S'$, and Δ_b^+ for $b \in S'$ are pairwise disjoint, we have $M_0 \subset \Omega$ and $M_1 \subset \Omega$. Moreover, M_0 and M_1 are compact smooth subsurfaces with boundary of Ω, and

$$M_0 \subset \operatorname{int} M_1 = B \backslash \bigcup_{b \in S'} \Delta_b^-.$$

The restriction $\pi^{-1}(\Omega) \to \Omega$ of π is a proper holomorphic submersion, so it is a smooth locally trivial fiber bundle by Ehresmann's theorem [Ehr51] (see [Voi02, Theorem 9.3], applied over coordinate disks in Ω). For any closed point $c \in \Omega$, $\pi^{-1}(c)$ is a rationally connected smooth projective variety (Construction-Proposition 3.2). In particular, $\pi^{-1}(c)$ is connected, and $\pi^{-1}(c)$ is simply connected by Lemma 2.6 and [DGX26, Theorem 3.1(ii)] (see also [Kol00, Proposition 2.3]).

Let s' be the smooth section of π over an open neighborhood of M_0 in Ω which is equal to u_b near $\Delta_b^+ \backslash \operatorname{int} \Delta_b$ for any $b \in S'$. By [DGX26, Lemma 2.3] (cf. [Ste51, §29]), applied to the smooth locally trivial fiber bundle $\pi^{-1}(\Omega) \to \Omega$ with connected and simply connected fibers, the compact smooth subsurfaces $M_0 \subset \operatorname{int} M_1$ of Ω, and the section s' over a neighborhood of M_0, there exists a smooth section s_1 of π over an open neighborhood of M_1 such that $s_1 = s'$ on an open neighborhood of M_0. Let $s_0 : B \to \mathcal{X}$ be the section of π which is equal to s_1 on

$$B \backslash \bigcup_{b \in S'} \operatorname{int} \Delta_b$$

and to u_b on Δ_b for any $b \in S'$. Since $s_1 = u_b$ on a neighborhood of $\partial \Delta_b \subset M_0$ for any $b \in S'$, s_0 is smooth, and $s_0|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$.

Finally, $s_0(\Delta_b) = u_b(\Delta_b) \subset \mathcal{X}^\circ$ for any $b \in S'$ (Construction-Proposition 3.3). Since π is smooth over Ω, we have $\pi^{-1}(\Omega) \subset \mathcal{X}^\circ$. Since

$$B \setminus \bigcup_{b \in S'} \operatorname{int} \Delta_b \subset M_1 \subset \Omega,$$

we have $s_0(c) \in \mathcal{X}^\circ$ for any $c \in B$ outside $\bigcup_{b\in S'} \operatorname{int} \Delta_b$. Thus $s_0(B) \subset \mathcal{X}^\circ$. □

4. Inserting rational curves

In this section, we prove Proposition 4.1, which raises the degree of $s_0^* T_{\mathcal{X}/B}$. Section 5 uses sections of large degree, and the sketch of the proof after Corollary 1.4 explains why the degree has to be raised.

Proposition 4.1. *Notation as in Construction-Proposition 3.3. Let $s_0 : B \to \mathcal{X}$ be a smooth section of π such that $s_0(B) \subset \mathcal{X}^\circ$ and $s_0|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$. Then there exists a positive integer a satisfying the following. For any integer $k \geq 0$, there exists a smooth section $s_k : B \to \mathcal{X}$ of π such that $s_k(B) \subset \mathcal{X}^\circ$, $s_k|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$, and $\deg s_k^* T_{\mathcal{X}/B} = \deg s_0^* T_{\mathcal{X}/B} + ka$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we trivialize π over a disk $\Delta' \subset \Omega$ and replace s_0 by a section s_0' such that $s_0'^* T_{\mathcal{X}/B}$ and $s_0^* T_{\mathcal{X}/B}$ have equal degree and the fiber coordinate of s_0' is constant near a point $c \in \Delta'$.

Since B is a closed surface, the disks Δ_b are finitely many and pairwise disjoint, and $B \setminus \Omega \subset S'$, the set $B \setminus \bigcup_{b\in S'} \Delta_b$ is a non-empty open subset of Ω. Choose an open subset Δ' of $B \setminus \bigcup_{b\in S'} \Delta_b$ and a holomorphic coordinate z mapping Δ' isomorphically onto $\{z \in \mathbb{C} : |z| < 2\}$. Let $c \in \Delta'$ be the point with $z(c) = 0$, and let $\mathcal{X}_c := \pi^{-1}(c)$. Since $\Delta' \subset \Omega$, we have $\pi^{-1}(\Delta') \subset \mathcal{X}^\circ$, and $\pi^{-1}(\Delta') \to \Delta'$ is a proper submersion. By Ehresmann's theorem [Ehr51] (see [Voi02, Theorem 9.3]), applied over the contractible manifold Δ' with base point c, there exists a diffeomorphism

$$\Phi : \pi^{-1}(\Delta') \to \Delta' \times \mathcal{X}_c$$

such that $\operatorname{pr}_1 \circ \Phi = \pi$. The map $\varphi := \operatorname{pr}_2 \circ \Phi|_{\mathcal{X}_c}$ is a diffeomorphism of $\mathcal{X}_c$, and $(\mathrm{id}_{\Delta'} \times \varphi^{-1}) \circ \Phi$ is again a diffeomorphism over Δ'. Possibly replacing Φ with $(\mathrm{id}_{\Delta'} \times \varphi^{-1}) \circ \Phi$, we may assume that

$$\Phi(y) = (c, y) \quad \text{for any } y \in \mathcal{X}_c. \tag{4.1}$$

Since the generic fiber of π is X and π is smooth over Ω, we have $\dim \mathcal{X}_c = d$. Let

$$\sigma := \operatorname{pr}_2 \circ \Phi \circ s_0|_{\Delta'} : \Delta' \to \mathcal{X}_c$$

and $x := \sigma(c)$. Choose an open neighborhood V of x in $\mathcal{X}_c$, a diffeomorphism ψ from V onto the open unit ball of $\mathbb{R}^{2d}$ with $\psi(x) = 0$, and a real number $r \in (0, 1]$ such that $\sigma(\{|z| \leq r\}) \subset V$. Let $\chi : \mathbb{R} \to [0, 1]$ be a smooth function with $\chi(t) = 0$ for $t \leq 1/3$ and $\chi(t) = 1$ for $t \geq 2/3$. Define $H : B \times [0, 1] \to \mathcal{X}$ by

$$H(p, \tau) := \begin{cases} \Phi^{-1}\Big(p, \psi^{-1}\big((1 - \tau + \tau\chi(|z(p)|/r))\psi(\sigma(p))\big)\Big) & \text{if } p \in \Delta' \text{ and } |z(p)| < r, \\ s_0(p) & \text{otherwise.} \end{cases}$$

Since the unit ball is convex and $0 \leq 1 - \tau + \tau\chi \leq 1$, H is well defined. Since $\chi(t) = 0$ for $t \leq 1/3$, the function $\chi(|z|/r)$ vanishes on $\{|z| \leq r/3\}$, so it is smooth on Δ', although $|z|$ is not smooth at $z = 0$. If $2r/3 \leq |z(p)| < r$, then $\chi(|z(p)|/r) = 1$, so $H(p, \tau) = \Phi^{-1}(p, \sigma(p)) = s_0(p)$ because $\operatorname{pr}_1 \Phi(s_0(p)) = \pi(s_0(p)) = p$. Thus H is smooth, $H(\cdot, \tau)$ is a section of π with image in $\mathcal{X}^\circ$ for any $\tau \in [0, 1]$, $H(\cdot, 0) = s_0$, and $H(p, \tau) = s_0(p)$ for any $p \in B$ with $p \notin \{|z| < 2r/3\}$ and any τ. Let $s_0' := H(\cdot, 1)$. Then $s_0'(B) \subset \mathcal{X}^\circ$, $s_0' = s_0$ on $B \setminus \{|z| < 2r/3\}$, in particular $s_0'|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$, and

$$\Phi(s_0'(p)) = (p, x) \quad \text{for any } p \in \Delta' \text{ with } |z(p)| \leq r/3. \tag{4.2}$$

Since H is a smooth homotopy through sections with image in $\mathcal{X}^\circ$, by [DGX26, Lemma 6.1] the complex vector bundles $s_0^*T_{\mathcal{X}/B}$ and $s_0'^*T_{\mathcal{X}/B}$ are smoothly isomorphic. Thus

$$\deg s_0'^*T_{\mathcal{X}/B} = \deg s_0^*T_{\mathcal{X}/B}. \tag{4.3}$$

Step 2. In this step we choose a very free rational curve $f : \mathbb{P}^1 \to \mathcal{X}_c$ with $f(\infty) = x$, and define a.

Since $c \in \Omega$, $\mathcal{X}_c$ is a rationally connected smooth projective variety (Construction-Proposition 3.2), hence rationally connected in the sense of [DGX26, §1.1] (Lemma 2.6). By [DGX26, Theorem 3.1(i)] (due to [Kol00, Definition–Theorem 2.1(5)]), applied to $\mathcal{X}_c$ with the single point $\infty \in \mathbb{P}^1$ and the jet of length 2 in the sense of [DGX26, §1.1] at ∞ of the constant map $\mathbb{P}^1 \to \{x\}$, that is, the 1-jet (Definition 2.3 with $N = 1$) with value x and first derivative zero, there exists a morphism $f : \mathbb{P}^1 \to \mathcal{X}_c$ such that $f(\infty) = x$ and

$$H^1\big(\mathbb{P}^1, f^*T_{\mathcal{X}_c}(-2[\infty])\big) = 0.$$

By [Gro57, Théorème 2.1] (see [Har77, Exercise V.2.6]),

$$f^*T_{\mathcal{X}_c} \cong \bigoplus_{i=1}^{d} \mathcal{O}_{\mathbb{P}^1}(a_i)$$

for some integers $a_1, \dots, a_d$. Then $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(a_i - 2)) = 0$ for any i, and [Har77, Theorem III.5.1] implies that $a_i - 2 \geq -1$, that is, $a_i \geq 1$. In particular, f is very free (Definition 2.5). Let $a := \deg f^*T_{\mathcal{X}_c}$, the degree of the underlying smooth complex vector bundle (Definition 2.7). By Lemma 2.8(1)(2),

$$a = \sum_{i=1}^{d} a_i \geq d \geq 1. \tag{4.4}$$

The integer a does not depend on k.

Step 3. In this step we construct s_k by inserting k copies of f into s_0'.

For $k = 0$, the section s_0 satisfies the conclusion. Assume that $k \geq 1$.

Let $\mathbb{D} := \{w \in \mathbb{C} : |w| \leq 1\}$, and let $\eta : \mathbb{R} \to [0, 1]$ be a smooth function with $\eta(t) = 1$ for $t \leq 1/16$, $\eta(t) > 0$ for $t < 1/4$, and $\eta(t) = 0$ for $t \geq 1/4$. Write $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$ and define $h : \mathbb{D} \to \mathbb{P}^1$ by

$$h(w) := \begin{cases} w/\eta(|w|^2) & \text{if } |w| < 1/2, \\ \infty & \text{if } |w| \geq 1/2. \end{cases}$$

On $\{|w| < 1/2\}$, h is a smooth map to $\mathbb{C}$. On $\{w \neq 0\}$, in the coordinate $1/\zeta$ of $\mathbb{P}^1$ near ∞, where ζ is the coordinate of $\mathbb{C} \subset \mathbb{P}^1$, h is the smooth function $w \to \eta(|w|^2)/w$. Thus h is smooth. Moreover, $h(w) = w$ for $|w| \leq 1/4$, and $h^{-1}(0) = \{0\}$.

Let S^2 be the sphere obtained by gluing two copies $\mathbb{D}_+$ and $\mathbb{D}_-$ of $\mathbb{D}$ along $\partial\mathbb{D}$ by the identity, oriented so that the inclusion of $\mathbb{D}_+$ preserves orientation and the inclusion of $\mathbb{D}_-$ reverses orientation. Let $\widetilde{h} : S^2 \to \mathbb{P}^1$ be h on $\mathbb{D}_+$ and ∞ on $\mathbb{D}_-$. Since h maps $\{|w| \geq 1/2\}$ to the point ∞, $\widetilde{h}$ is continuous. The point $0 \in \mathbb{P}^1$ has the single preimage $0 \in \mathbb{D}_+$, near which $\widetilde{h}$ is the identity. Thus $\widetilde{h}$ has degree one, that is,

$$\widetilde{h}_*[S^2] = [\mathbb{P}^1], \tag{4.5}$$

where $\mathbb{P}^1$ carries the orientation induced by the complex structure.

Choose points $z_1, \dots, z_k \in \mathbb{C}$ and a real number $\rho > 0$ such that the closed disks

$$D_j := \{p \in \Delta' : |z(p) - z_j| \leq \rho\}, \quad 1 \leq j \leq k,$$

are pairwise disjoint and contained in $\{|z| < r/3\}$. Define $s_k : B \to \mathcal{X}$ by

$$s_k(p) := \begin{cases} \Phi^{-1}\Big(p, f\big(h((z(p) - z_j)/\rho)\big)\Big) & \text{if } p \in D_j \text{ for some } 1 \le j \le k, \\ s_0'(p) & \text{otherwise.} \end{cases}$$

If $p \in D_j$ and $|z(p) - z_j| \ge \rho/2$, then $s_k(p) = \Phi^{-1}(p, f(\infty)) = \Phi^{-1}(p, x) = s_0'(p)$ by (4.2). Thus s_k is a smooth section of π, and $s_k = s_0'$ on $B \setminus \bigcup_{j=1}^k D_j$. Since $\bigcup_{j=1}^k D_j \subset \Delta'$, $\pi^{-1}(\Delta') \subset \mathcal{X}^\circ$, and $\Delta' \cap \Delta_b = \emptyset$ for any $b \in S'$, we have $s_k(B) \subset \mathcal{X}^\circ$ and $s_k|_{\Delta_b} = s_0'|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$.

Step 4. In this step we prove that $\deg s_k^* T_{\mathcal{X}/B} = \deg s_0^* T_{\mathcal{X}/B} + ka$.

For any $1 \le j \le k$, let $e_j : \mathbb{D} \to D_j$ be the orientation-preserving diffeomorphism with $z(e_j(w)) = z_j + \rho w$, and let $g_j : S^2 \to \pi^{-1}(\Delta')$ be $s_k \circ e_j$ on $\mathbb{D}_+$ and $s_0' \circ e_j$ on $\mathbb{D}_-$. Since $s_k = s_0'$ near ∂D_j, g_j is continuous. In what follows, $g_{j*}[S^2]$ denotes the pushforward of $[S^2]$ under the composite of g_j and the inclusion $\pi^{-1}(\Delta') \subset \mathcal{X}^\circ$. Choose a singular 2-cycle β representing $[B]$ of the form $\beta = \gamma + \sum_{j=1}^k e_{j*}[\mathbb{D}]$, where $[\mathbb{D}]$ is a relative fundamental chain of $\mathbb{D}$ and γ is a 2-chain in $B \setminus \bigcup_{j=1}^k \operatorname{int} D_j$ (such β exists, for instance the sum of the oriented 2-simplices of a triangulation of B whose restriction to each D_j is the image under e_j of a fixed triangulation of $\mathbb{D}$). Since $s_k = s_0'$ on $B \setminus \bigcup_{j=1}^k \operatorname{int} D_j$, we have

$$s_{k*}\beta - s_{0*}'\beta = \sum_{j=1}^k \big((s_k \circ e_j)_*[\mathbb{D}] - (s_0' \circ e_j)_*[\mathbb{D}]\big),$$

and the j-th summand is a cycle representing $g_{j*}[S^2]$ because the inclusion of $\mathbb{D}_-$ reverses orientation. Thus in $H_2(\mathcal{X}^\circ; \mathbb{Z})$,

$$s_{k*}[B] = s_{0*}'[B] + \sum_{j=1}^k g_{j*}[S^2]. \tag{4.6}$$

Let $\iota_c : \mathcal{X}_c \to \mathcal{X}^\circ$ be the inclusion. By (4.1), the map

$$\pi^{-1}(\Delta') \times [0,1] \to \pi^{-1}(\Delta'), \quad (y, \tau) \to \Phi^{-1}\Big(z^{-1}\big((1-\tau)z(\pi(y))\big), \operatorname{pr}_2 \Phi(y)\Big)$$

is a homotopy from the identity of $\pi^{-1}(\Delta')$ to the composite of $\operatorname{pr}_2 \circ \Phi$, viewed as a map $\pi^{-1}(\Delta') \to \mathcal{X}_c$, and the inclusion $\mathcal{X}_c \subset \pi^{-1}(\Delta')$. Composing with the inclusion $\pi^{-1}(\Delta') \subset \mathcal{X}^\circ$, the inclusion $\pi^{-1}(\Delta') \subset \mathcal{X}^\circ$ is homotopic to $\iota_c \circ \operatorname{pr}_2 \circ \Phi$. By the definitions of s_k and g_j and by (4.2), $\operatorname{pr}_2 \circ \Phi \circ g_j = f \circ \widetilde{h}$. Thus, by (4.5), in $H_2(\mathcal{X}^\circ; \mathbb{Z})$,

$$g_{j*}[S^2] = \iota_{c*}(\operatorname{pr}_2 \circ \Phi \circ g_j)_*[S^2] = \iota_{c*} f_* \widetilde{h}_*[S^2] = \iota_{c*} f_*[\mathbb{P}^1] \tag{4.7}$$

for any $1 \le j \le k$. In particular, $g_{j*}[S^2]$ does not depend on j.

By naturality of Chern classes [MS74, Lemma 14.2], for any smooth section s of π with $s(B) \subset \mathcal{X}^\circ$,

$$\deg s^* T_{\mathcal{X}/B} = \langle c_1(s^* T_{\mathcal{X}/B}), [B] \rangle = \langle c_1(T_{\mathcal{X}/B}), s_*[B] \rangle. \tag{4.8}$$

Since $\iota_c^* T_{\mathcal{X}/B} = T_{\mathcal{X}_c}$, naturality of Chern classes [MS74, Lemma 14.2] and Definition 2.7 imply that

$$\langle c_1(T_{\mathcal{X}/B}), \iota_{c*} f_*[\mathbb{P}^1] \rangle = \langle c_1(T_{\mathcal{X}_c}), f_*[\mathbb{P}^1] \rangle = \langle c_1(f^* T_{\mathcal{X}_c}), [\mathbb{P}^1] \rangle = \deg f^* T_{\mathcal{X}_c} = a. \tag{4.9}$$

By (4.8), (4.6), (4.7), (4.9), and (4.3),

$$\begin{aligned} \deg s_k^* T_{\mathcal{X}/B} &= \langle c_1(T_{\mathcal{X}/B}), s_{0*}'[B] \rangle + \sum_{j=1}^k \langle c_1(T_{\mathcal{X}/B}), \iota_{c*} f_*[\mathbb{P}^1] \rangle \\ &= \deg s_0'^* T_{\mathcal{X}/B} + ka = \deg s_0^* T_{\mathcal{X}/B} + ka. \end{aligned} \tag{4.10}$$

Together with **Step 3**, s_k satisfies the conclusion, and a is a positive integer by (4.4). □

5. Proof of the main theorem

In this section, we construct the bundle data in Theorem 2.11 (Proposition 5.1), prove Proposition 5.2, and deduce Theorem 1.2 and Corollary 1.4.

Proposition 5.1. *Notation as in Construction-Proposition 3.3. Let $s : B \to \mathcal{X}$ be a smooth section of π such that $s(B) \subset \mathcal{X}^\circ$, $s|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$, and $\deg s^*T_{\mathcal{X}/B} \geq d \deg D + 2gd$. Let $\mathcal{E}_s$ be as in Construction 2.10. Then there exist a holomorphic vector bundle $\mathcal{V}$ on B with $H^1(B, \mathcal{V}) = 0$, a smooth complex vector bundle isomorphism $\phi : \mathcal{V} \to \mathcal{E}_s(-D)$, and an open subset $U \subset B$ containing S', such that s and ϕ are holomorphic on U.*

Proof. Let $e := \deg \mathcal{E}_s(-D)$. Since the underlying smooth complex vector bundle of $\mathcal{E}_s$ is $s^*T_{\mathcal{X}/B}$ and $\operatorname{rank} s^*T_{\mathcal{X}/B} = d$, Lemma 2.8(1)(2) implies that

$$e = \deg s^*T_{\mathcal{X}/B} - d \deg D \geq 2gd.$$

Write $e = dq + r$ with integers q and r such that $0 \leq r < d$. Since $dq = e - r > 2gd - d$, we have $q \geq 2g$. Fix a closed point $b_0 \in B$ and let

$$\mathcal{V} := \mathcal{O}_B((q+1)b_0)^{\oplus r} \oplus \mathcal{O}_B(qb_0)^{\oplus(d-r)}.$$

Since

$$\deg \mathcal{O}_B((q+1)b_0) = q + 1 > 2g - 2 \quad \text{and} \quad \deg \mathcal{O}_B(qb_0) = q > 2g - 2,$$

we have $H^1(B, \mathcal{V}) = 0$ by [Har77, Example IV.1.3.4] and Serre duality [Har77, Corollary III.7.7] (the holomorphic and algebraic cohomology of $\mathcal{V}$ coincide by [Har77, Appendix B, Theorem 2.1]). Moreover, $\operatorname{rank} \mathcal{V} = d = \operatorname{rank} \mathcal{E}_s(-D)$, and Lemma 2.8(1)(2) implies that

$$\deg \mathcal{V} = r(q+1) + (d-r)q = e = \deg \mathcal{E}_s(-D).$$

By Lemma 2.8(3), there exists a smooth complex vector bundle isomorphism $\phi_0 : \mathcal{V} \to \mathcal{E}_s(-D)$.

For any $b \in S'$, let $\Delta'_b \subset \operatorname{int} \Delta_b$ be a closed coordinate disk centered at b, and let $P := \bigcup_{b \in S'} \Delta'_b$. Since S' is finite, there exists a closed coordinate disk $\Delta^* \subset B$ such that $\Delta^* \cap \Delta_b = \emptyset$ for any $b \in S'$. Let $W := B \backslash \Delta^*$. Then W is a connected non-compact open Riemann surface, relatively compact in B, and $P \subset W$. Thus W is an open Riemann surface in the sense of [DGX26, §1.1]. Recall that, for holomorphic vector bundles V and E of equal rank on a compact Riemann surface, a smooth complex vector bundle isomorphism $V \to E$, and a compact subset contained in a relatively compact open Riemann surface, [DGX26, Lemma 6.2], in the case where the second compact set is empty, implies that there exists a smooth complex vector bundle isomorphism $V \to E$ which is holomorphic on a neighborhood of the compact subset. Applying it to ϕ_0 and $P \subset W$, we obtain a smooth complex vector bundle isomorphism $\phi : \mathcal{V} \to \mathcal{E}_s(-D)$ which is holomorphic on an open neighborhood U_0 of P.

Let

$$U := U_0 \cap \bigcup_{b \in S'} \operatorname{int} \Delta'_b.$$

Then U is an open subset of B containing S', and ϕ is holomorphic on U. For any $b \in S'$, since $s|_{\Delta_b} = u_b|_{\Delta_b}$ and u_b is holomorphic on an open neighborhood of Δ_b, s is holomorphic on $\operatorname{int} \Delta_b$. Since

$$U \cap \operatorname{int} \Delta'_b \subset \operatorname{int} \Delta_b$$

for any $b \in S'$, s is holomorphic on U. □

Proposition 5.2. *Notation as in Construction-Proposition 3.3. Then there exists a section $\sigma : B \to \mathcal{X}$ of π such that $j_b^{N_b}\sigma = j_b^{N_b}\hat{x}_b$ for any $b \in S$.*

Proof. By Proposition 3.4, there exists a smooth section $s_0 : B \to \mathcal{X}$ of π such that $s_0(B) \subset \mathcal{X}^\circ$ and $s_0|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$. Let a be the positive integer in Proposition 4.1 for s_0. Since $a > 0$, there exists an integer $k \geq 0$ such that

$$\deg s_0^*T_{\mathcal{X}/B} + ka \geq d \deg D + 2gd.$$

Let $s := s_k$, where s_k is the smooth section of π in Proposition 4.1. Then $s(B) \subset \mathcal{X}^\circ$, $s|_{\Delta_b} = u_b|_{\Delta_b}$ for any $b \in S'$, and $\deg s^*T_{\mathcal{X}/B} \geq d \deg D + 2gd$. By Proposition 5.1, there exist a holomorphic vector bundle $\mathcal{V}$ on B with $H^1(B, \mathcal{V}) = 0$, a smooth complex vector bundle isomorphism $\phi : \mathcal{V} \to \mathcal{E}_s(-D)$, and an open subset $U \subset B$ containing S', such that s and ϕ are holomorphic on U.

We check the hypotheses of Theorem 2.11 for π, Ω, s, S, N_b, D, U, $\mathcal{V}$, and ϕ.

(1) Since $\mathcal{X}$ is a smooth projective variety, $\mathcal{X}$ is a complex manifold. Since π is a projective morphism, π is a proper holomorphic map.
(2) Since Ω is a Zariski open subset of B, Ω is open in B. Since π is smooth over Ω, $\pi^{-1}(\Omega) \to \Omega$ is a holomorphic submersion. Let $\iota : \mathcal{X} \to \mathbb{P}^m$ be a closed embedding. Since B is separated, $(\pi, \iota) : \mathcal{X} \to B \times \mathbb{P}^m$ is a closed embedding over B. Thus $\pi^{-1}(\Omega) \to \Omega$ is a smooth projective morphism in the sense of [DGX26, Definition 1.1]. By Construction-Proposition 3.2, $\pi^{-1}(c)$ is a rationally connected smooth projective variety for any $c \in \Omega$, hence $\pi^{-1}(c)$ is rationally connected in the sense of [DGX26, §1.1] (Lemma 2.6). Since [DGX26, Proposition 3.5] holds for any smooth projective morphism in the sense of [DGX26, Definition 1.1] whose base is a complex manifold, $\pi^{-1}(\Omega) \to \Omega$ has the relative first-jet sphere-family property.
(3) The map s is a smooth section of π, and $s(B) \subset \mathcal{X}^\circ$, where $\mathcal{X}^\circ$ is the submersion locus of π.
(4) The set S is finite, $N_b \geq 0$ is an integer for any $b \in S$, and $D = \sum_{b \in S}(N_b + 1)b$.
(5) The open subset U contains $S' = S \cup (B \backslash \Omega)$, and s and ϕ are holomorphic on U.
(6) The bundle $\mathcal{V}$ is a holomorphic vector bundle on B with $H^1(B, \mathcal{V}) = 0$, and $\phi : \mathcal{V} \to \mathcal{E}_s(-D)$ is a smooth complex vector bundle isomorphism.

By Theorem 2.11, there exists a holomorphic section $\sigma : B \to \mathcal{X}$ of π such that $j_b^{N_b}\sigma = j_b^{N_b}s$ for any $b \in S$, in the sense of Definition 2.3(2).

For holomorphic maps from a neighborhood of b in B to the smooth variety $\mathcal{X}$, the conditions in Definition 2.3(1) and Definition 2.3(2) are equivalent, since both say that, for a local coordinate t of B centered at b, the power series expansions in t of the maps coincide up to the term t^N (for (1), the completions of the algebraic and analytic local rings of $\mathcal{X}$ at the common value of the maps at b coincide [Ser56, Proposition 3]). In particular, $j_b^{N_b}\sigma = j_b^{N_b}s$ also holds in the sense of Definition 2.3(1). For any $b \in S$, since $s|_{\Delta_b} = u_b|_{\Delta_b}$, Construction-Proposition 3.3 implies that

$$j_b^{N_b}\sigma = j_b^{N_b}s = j_b^{N_b}u_b = j_b^{N_b}\hat{x}_b.$$

It remains to show that σ is a morphism. The graph $\Gamma \subset B \times \mathcal{X}$ of σ is a closed complex submanifold of dimension one of the projective variety $B \times \mathcal{X}$, and the first projection restricts to a biholomorphic map from Γ to the connected Riemann surface B. Thus Γ is an irreducible algebraic curve by [Cho49, Theorem V] (Γ is connected and smooth, hence irreducible). The projection $p : \Gamma \to B$ is a bijective morphism, hence birational in characteristic zero (by generic smoothness [Har77, Corollary III.10.7], applied to the normalization of Γ, a general fiber of p consists of $[K(\Gamma) : K(B)]$ points). Let $\nu : \widetilde{\Gamma} \to \Gamma$ be the normalization. Then $p \circ \nu$ is a birational morphism between smooth projective curves ($\widetilde{\Gamma}$ is projective by [Har77, Proposition II.6.7]), hence an isomorphism by [Har77, Corollary I.6.12]. Thus $\sigma = \mathrm{pr}_2 \circ \nu \circ (p \circ \nu)^{-1}$ is a morphism, where $\mathrm{pr}_2 : B \times \mathcal{X} \to \mathcal{X}$ is the second projection. □

Proof of Theorem 1.2. Let X be a smooth proper geometrically rationally connected variety over F. By [HT06, §2.1], there exists a smooth projective variety X' over F birational to X. Since X and X' are proper, X' is geometrically rationally connected [Kol96, Proposition IV.3.3, (3.3.3)]. Since weak approximation is a birational property of smooth varieties [HT06, §2.1], possibly replacing X with X', we may assume that X is projective.

The case $\dim X = 0$ is trivial. Thus we may assume that $\dim X \geq 1$. Let S be a finite set of closed points of B. For any $b \in S$, let $x_b \in X(F_b)$ and let $\mathcal{U}_b$ be an open neighborhood of x_b in

$X(F_b)$. It suffices to show that there exists $x \in X(F)$ whose image in $X(F_b)$ belongs to $\mathcal{U}_b$ for any $b \in S$.

Let g be the genus of B and $d := \dim X$. Then the conditions in Set-up 3.1(1) and (2) hold, and we fix the model $\pi : \mathcal{X} \to B$ of X and the Zariski open subset $\Omega \subset B$ of Construction-Proposition 3.2. By Lemma 2.4 applied to π, for any $b \in S$, there exists an integer $N_b \geq 0$ such that

$$\{y \in X(F_b) \mid j_b^{N_b} \hat{y} = j_b^{N_b} \hat{x}_b\} \subset \mathcal{U}_b,$$

where $\hat{x}_b, \hat{y} : \widehat{B}_b \to \mathcal{X}$ are the extensions of x_b and y respectively. Let $D := \sum_{b \in S}(N_b + 1)b$. Then the conditions of Set-up 3.1 hold. We fix Δ_b and u_b for $b \in S'$ as in Construction-Proposition 3.3. By Proposition 5.2, there exists a section $\sigma : B \to \mathcal{X}$ of π such that $j_b^{N_b}\sigma = j_b^{N_b}\hat{x}_b$ for any $b \in S$.

Since the generic fiber of π is X, the restriction of σ to the generic point of B induces a point $x \in X(F)$. For any $b \in S$, by the uniqueness in Lemma 2.4, the extension $\hat{x} : \widehat{B}_b \to \mathcal{X}$ of the image of x in $X(F_b)$ is the restriction of σ to $\widehat{B}_b$. Thus $j_b^{N_b}\hat{x} = j_b^{N_b}\sigma = j_b^{N_b}\hat{x}_b$, hence the image of x in $X(F_b)$ belongs to $\mathcal{U}_b$. □

Proof of Corollary 1.4. For any $b \in S$, the restriction of $\hat{x}_b$ to the generic point of $\widehat{B}_b$ induces a point $x_b \in X(F_b)$, and $\hat{x}_b$ is the extension of x_b by the uniqueness in Lemma 2.4. By Theorem 1.2 and Lemma 2.4, there exists $x \in X(F)$ such that $j_b^{N_b}\hat{x} = j_b^{N_b}\hat{x}_b$ for any $b \in S$, where $\hat{x} : \widehat{B}_b \to \mathcal{X}$ is the extension of the image of x in $X(F_b)$.

Let $Z \subset \mathcal{X}$ be the closure of x, where x is regarded as a point of the generic fiber of π. Then Z is an irreducible curve, and $Z \to B$ is proper and birational, since the residue field of x is F. Let $\nu : \widetilde{Z} \to Z$ be the normalization. Then $\widetilde{Z}$ is a complete smooth curve, hence projective [Har77, Proposition II.6.7], and $\widetilde{Z} \to B$ is a birational morphism between smooth projective curves, hence an isomorphism [Har77, Corollary I.6.12]. Let $\sigma : B \to \mathcal{X}$ be the composition of the inverse of $\widetilde{Z} \to B$ with ν and the inclusion $Z \subset \mathcal{X}$. Then σ is a section of π. For any $b \in S$, the restriction of σ to $\widehat{B}_b$ is $\hat{x}$ by the uniqueness in Lemma 2.4. Thus $j_b^{N_b}\sigma = j_b^{N_b}\hat{x} = j_b^{N_b}\hat{x}_b$. □

References

[AF25] A. Alarcón and F. Forstnerič, *Oka-1 manifolds*, Math. Z. **311** (2025), no. 2.

[BW21] O. Benoist and O. Wittenberg, *The tight approximation property*, J. Reine Angew. Math. **776** (2021), 151–200.

[BW25] O. Benoist and O. Wittenberg, *Tight approximation for rationally simply connected varieties*, Taiwanese J. Math. **29** (2025), no. 6, 1165–1183.

[Cam92] F. Campana, *Connexité rationnelle des variétés de Fano*, Ann. Sci. École Norm. Sup. (4) **25** (1992), no. 5, 539–545.

[Cho49] W.-L. Chow, *On compact complex analytic varieties*, Amer. J. Math. **71** (1949), no. 4, 893–914.

[CTG04] J.-L. Colliot-Thélène and P. Gille, *Remarques sur l'approximation faible sur un corps de fonctions d'une variable*, in: *Arithmetic of higher-dimensional algebraic varieties (Palo Alto, CA, 2002)*, Progr. Math. **226**, Birkhäuser Boston, Boston, MA (2004), 121–134.

[Dan26] S. Dandapat, *Campana rational connectedness and weak approximation of del Pezzo orbifolds*, arXiv:2609.11533.

[dJHS11] A. J. de Jong, X. He, and J. M. Starr, *Families of rationally simply connected varieties over surfaces and torsors for semisimple groups*, Publ. Math. Inst. Hautes Études Sci. **114** (2011), 1–85.

[DGX26] Y.-H. Du, B. Guo, and S.-Y. Xie, *Analytic and algebraic Oka-1 approximation for smooth projective morphisms with rationally connected fibers*, arXiv:2609.11883.

[Ehr51] C. Ehresmann, *Les connexions infinitésimales dans un espace fibré différentiable*, in: *Colloque de topologie (espaces fibrés), Bruxelles, 1950*, Georges Thone, Liège; Masson et Cie, Paris (1951), 29–55.

[FL25] F. Forstnerič and F. Lárusson, *Oka-1 manifolds: new examples and properties*, Math. Z. **309** (2025), no. 2, Paper No. 26.

[GHS03] T. Graber, J. Harris, and J. M. Starr, *Families of rationally connected varieties*, J. Amer. Math. Soc. **16** (2003), no. 1, 57–67.

[GH78] P. Griffiths and J. Harris, *Principles of algebraic geometry*, Pure and Applied Mathematics, Wiley-Interscience, New York (1978).

[Gro57] A. Grothendieck, *Sur la classification des fibrés holomorphes sur la sphère de Riemann*, Amer. J. Math. **79** (1957), 121–138.

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, New York (1977).

[Has10] B. Hassett, *Weak approximation and rationally connected varieties over function fields of curves*, in: *Variétés rationnellement connexes: aspects géométriques et arithmétiques*, Panor. Synthèses **31**, Soc. Math. France, Paris (2010), 115–153.

[HT06] B. Hassett and Y. Tschinkel, *Weak approximation over function fields*, Invent. Math. **163** (2006), no. 1, 171–190.

[HT08] B. Hassett and Y. Tschinkel, *Approximation at places of bad reduction for rationally connected varieties*, Pure Appl. Math. Q. **4** (2008), no. 3, 743–766.

[HT09] B. Hassett and Y. Tschinkel, *Weak approximation for hypersurfaces of low degree*, in: *Algebraic geometry—Seattle 2005. Part 2*, Proc. Sympos. Pure Math. **80**, Amer. Math. Soc., Providence, RI (2009), 937–955.

[Hir64] H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II*, Ann. of Math. (2) **79** (1964), 109–203; 205–326.

[Kne13] A. Knecht, *Weak approximation for general degree two del Pezzo surfaces*, Proc. Amer. Math. Soc. **141** (2013), no. 3, 801–811.

[Kol96] J. Kollár, *Rational curves on algebraic varieties*, Ergeb. Math. Grenzgeb. (3) **32**, Springer-Verlag, Berlin (1996).

[Kol00] J. Kollár, *Fundamental groups of rationally connected varieties*, Michigan Math. J. **48** (2000), no. 1, 359–368.

[Kol07] J. Kollár, *Lectures on resolution of singularities*, Ann. of Math. Stud. **166**, Princeton University Press, Princeton, NJ (2007).

[KMM92] J. Kollár, Y. Miyaoka, and S. Mori, *Rationally connected varieties*, J. Algebraic Geom. **1** (1992), no. 3, 429–448.

[MS74] J. W. Milnor and J. D. Stasheff, *Characteristic classes*, Annals of Mathematics Studies, No. 76, Princeton University Press, Princeton, NJ (1974).

[RS09] M. Roth and J. M. Starr, *A local-global principle for weak approximation on varieties over function fields*, arXiv:0908.0096.

[Ser56] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier (Grenoble) **6** (1956), 1–42.

[Sta06] J. M. Starr, *Hypersurfaces of low degree are rationally simply-connected*, arXiv:math/0602641.

[Ste51] N. Steenrod, *The topology of fibre bundles*, Princeton Math. Ser. **14**, Princeton University Press, Princeton, NJ (1951).

[Tia15] Z. Tian, *Weak approximation for cubic hypersurfaces*, Duke Math. J. **164** (2015), no. 7, 1401–1435.

[TZ19] Z. Tian and R. Zong, *Weak approximation for isotrivial families*, J. Reine Angew. Math. **752** (2019), 1–23.

[Voi02] C. Voisin, *Hodge theory and complex algebraic geometry. I*, Cambridge Stud. Adv. Math. **76**, Cambridge University Press, Cambridge (2002).

[Xu12] C. Xu, *Weak approximation for low degree del Pezzo surfaces*, J. Algebraic Geom. **21** (2012), no. 4, 753–767.

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